Pandas to PySpark

Pandas Method	PySpark Method	Use of Method
pd.read_csv()	spark.read.csv()	Read a CSV file into a DataFrame.
pd.read_excel()	spark.read.format("excel").load()	Read an Excel file into a DataFrame.
pd.read_json()	spark.read.json()	Read a JSON file or string into a DataFrame.
pd.Series()	pyspark.sql.Column / Single-col DataFrame	Create a 1D column/series equivalent data structure.
pd.DataFrame	spark.createDataFrame()	Create a 2D labeled data structure (table).
df.columns	df.columns	Return column names from the DataFrame.
df.shape	(df.count(), len(df.columns))	Return total number of rows and columns.
df.values	df.collect() / df.rdd	Extract underlying row data array.
df.info()	df.printSchema()	Display schema, column names, and data types.
df.head()	df.show() / df.limit()	Return top N rows (default 5).
df.tail()	df.tail()	Return bottom N rows (default 5).
df.describe()	df.describe() / df.summary()	Compute summary statistics for numeric columns.
df.fillna()	df.fillna() / df.na.fill()	Fill missing/null values (NaN, None) with a value.
df.dropna()	df.dropna() / df.na.drop()	Remove rows/columns containing missing values.
df.astype()	df.withColumn(col, col.cast())	Convert or cast column data types.
df.drop()	df.drop()	Delete specified columns from the DataFrame.
df.rename()	df.withColumnRenamed()	Rename column headers.
df.replace()	df.replace() / F.when()	Substitute old specific values with new values.
df.drop_duplicates()	df.dropDuplicates() / df.distinct()	Remove duplicate rows from the DataFrame.
df.duplicated()	df.groupBy(cols).count().filter("count > 1")	Detect and identify duplicate rows.
pd.to_datetime()	F.to_timestamp() / F.to_date()	Convert/parse string columns into datetime or date format.
pd.cut()	Bucketizer / F.when()	Bin continuous data into discrete intervals.

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`df.map()`	`F.transform() / PySpark UDF`	Apply element-wise mapping or custom function transformations.
`df.sort_values()`	`df.sort() / df.orderBy()`	Sort DataFrame rows by values in specified columns.
`pd.merge()`	`df1.join(df2)`	Merge/join two DataFrames using database-style join operations.
`pd.concat()`	`df1.union() / df1.unionByName()`	Concatenate two DataFrames vertically along rows.
`pd.pivot() / pd.pivot_table()`	`df.groupBy().pivot().agg()`	Reshape data into a pivot layout with aggregation.
`pd.melt()`	`df.selectExpr("stack(...)")`	Unpivot a DataFrame from wide-format to long-format.
`df.groupby()`	`df.groupBy()`	Group data based on column values for aggregation.
`df.agg()`	`df.agg()`	Apply one or more aggregate functions to a DataFrame.
`df.isnull()`	`F.col().isNull()`	Check for null or missing values within data columns.
`df[col].unique()`	`df.select('col').distinct()`	Fetch unique values present in a column.
`df[col].nunique()`	`df.select(F.countDistinct('col'))`	Count total number of unique values in a column.
`df[col].value_counts`	`df.groupBy('col').count()`	Calculate occurrences/frequency of distinct values in a column.
`count()`	`df.count() / F.count()`	Count non-missing values or total rows.
`sum()`	`F.sum()`	Return the sum of numeric values in a column.
`mean()`	`F.avg() / F.mean()`	Compute the average/mean of numerical values in a column.
`min()`	`F.min()`	Find the minimum scalar value in a column.
`max()`	`F.max()`	Find the maximum scalar value in a column.
`idxmax()`	`F.max_by() / Window`	Find the row value/descriptor associated with the maximum value.
`idxmin()`	`F.min_by() / Window`	Find the row value/descriptor associated with the minimum value.
`df.corr()`	`df.stat.corr()`	Compute pairwise correlation between numeric columns.
`Str . (built-in methods)`	`F.lower(), F.substring(), etc.`	Perform string manipulations on column data.